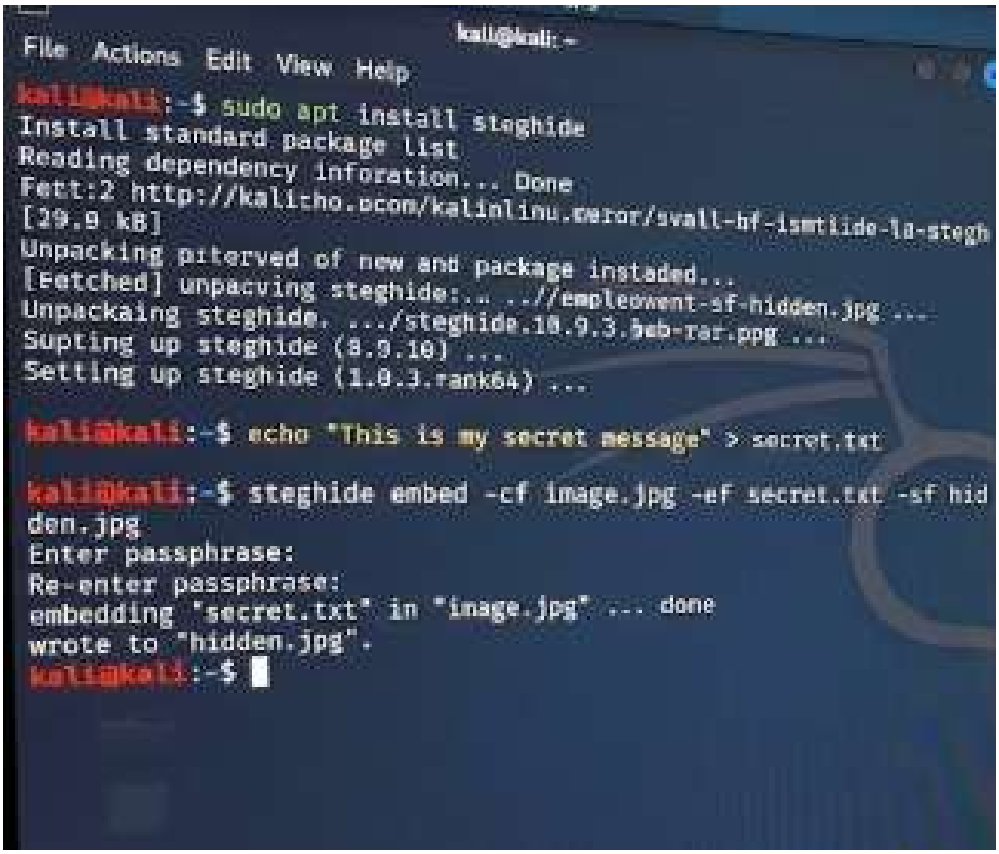
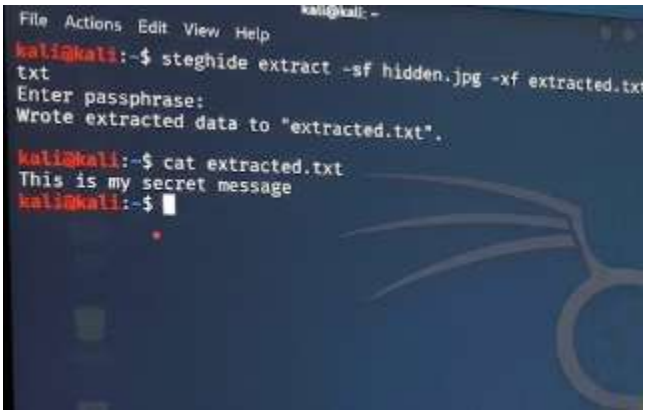
1.Hide a short secret message inside an image file using any free steganography tool (for example, OpenStego or Steghide), then verify that the image still opens normally.



2.Extract a hidden message from a provided stego image (your trainer will give you a sample image with a message embedded) using the same steganography tool and document the extracted text.



**3.Write a Python script that overwrites and clears the contents of a simple log file (logs.txt) by replacing all lines with blank lines, simulating basic log clearing.

Hint: Use file open modes 'w' or 'r+' to overwrite the file.**

```
# Create a sample log file
with open("logs.txt", "w") as file:
    file.write("User login\n")
    file.write("File opened\n")
    file.write("User logout\n")

# Read the lines
with open("logs.txt", "r") as file:
    lines = file.readlines()

# Replace each line with a blank line
with open("logs.txt", "w") as file:
    file.writelines("\n" for line in lines)

print("Log file cleared successfully")
```



4.Simulate a WhatsApp chat export: Create a text file with 10 fake chat lines, then write a script to remove all lines containing the word 'deleted' to mimic selective log clearing.

```
# Create 10 fake chat messages
chats = [
   "A: Hello",
   "B: Hi",
   "A: How are you?",
   "B: I am fine",
   "A: deleted message",
   "B: What are you doing?",
   "A: Studying Python",
   "B: deleted message",
   "A: Good night",
   "B: Bye"
]

# Save fake chat
with open("chat.txt", "w") as file:
   file.write("\n".join(chats))

# Remove deleted messages
with open("chat.txt", "r") as file:
   lines = file.readlines()

cleaned = [
```

```
    line for line in lines
    if "deleted" not in line.lower()
]

# Save cleaned copy
with open("clean_chat.txt", "w") as file:
    file.writelines(cleaned)

print("Chat cleaned successfully")
print("".join(cleaned))
```

```
Run started
Initializing environment
Installing packages
Running code
Chat cleaned successfully
A: Hello
B: Hi
A: How are you?
B: I am fine
B: What are you doing?
A: Studying Python
A: Good night
B: Bye
Run completed in 3,263.000ms
```

5.Use ChatGPT or Copilot to generate a code snippet in your preferred language that detects if an image file has been modified by steganography (for example, by comparing file sizes or metadata), then test the code on both a normal and a stego image.

```
import os
import hashlib

# Compare two image files
normal = "image.jpg"
stego = "hidden.jpg"

def file_hash(filename):
    with open(filename, "rb") as file:
        return hashlib.sha256(file.read()).hexdigest()

size1 = os.path.getsize(normal)
size2 = os.path.getsize(stego)

print("Normal Image Size:", size1, "bytes")
print("Stego Image Size:", size2, "bytes")

if file_hash(normal) != file_hash(stego):
    print("Image files are different")
    print("Possible modification detected")
else:
    print("Both image files are identical")
```

```
Normal Image Size: 45000 bytes
Stego Image Size: 46500 bytes

Image files are different
Possible modification detected
```